

AC-3 – NibbleAI
AI 7993 – Section W01 – Spring 2025
Jan 26, 2025

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 Ryan Deem	

Team Members:

Name	Role	Cell Phone / Alt Email
Zeynep (Team Lead)	Developer	(404)-906-2654 zbirgili@students.kennesaw.edu zeynepbirgili@outlook.com
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Overview / Abstract:

This Typescript based phone application will aim to discover recipes for the users based on available ingredients in their fridge or pantry using various AI object recognition tools. There will be 2 primary datasets, one dataset for all ingredient images, and another for all recipes. This will involve developing a food item recipe matching algorithm.

Platform:

The phone application will be built on React Native or Flutter frameworks. This will ensure that we are using our time effectively without learning high-level languages such as Swift or Kotlin and prioritizing TypeScript. The UI/UX will be designed on Figma. The application will be demonstrated on Expo Go. It will have a server side to tie the backend to the frontend and to store the learnt model.

We plan to use Python for the backend to allow us to run an ML model that we have trained. Then the AI framework in Python we will be using TensorFlow to create at least a MobileNet Network to identify each ingredient in an image. Then we feed the ingredient lookup table for possible recipes.

Collaboration Tools:

We will have weekly and then closer to the due date bi-weekly meetings.

- Collaboration - Our codebase and Kanban board are uploaded on GitHub.
- Communication - Discord and Text/Cell.

1/26/2025

Project website link(s)

[Work in Progress]

- <https://nibbleai.netlify.app/> (will be the main domain)
- <https://zbirdgilli.github.io/Smart-Recipe-Assistant/> (currently working project website)

Deliverables:

- React Native project will be delivered in .zip folder for an app worth 3-4 pages. This will include a navigation bar and various dependencies to support python scripts for our ML model.
- The ML model will be hosted via Microsoft Azure so we can seamlessly integrate it into our TypeScript/CSS/HTML UI and React platform to host on iOS and Android.

Milestone Events:

- Finishing Figma/Canva designs and building the UI
- Integrate UI on to React project structure
- Deploying the ML structure into the React project base
- Build a machine vision script to accurately identify objects
- Building highly accurate object to food model

STATEMENT OF PARTICIPATION (all team members must sign this, and scan it into documents)

Final Report Package (Group Assignment)

- Final Report (Group Assignment)
- Source Code (Group Assignment)
- Website (Group Assignment)
- Video Demo (Group Assignment)
- iOS and Android compatible mobile Recipe app
- C-Day Application/Submission

By signing below, I _____ (print name clearly) acknowledge that I will participate in all meetings, communications, deliverables and other tasks necessary to complete the project. If I do not participate, I understand that the course instructor will meet with me to remedy the situation.

_____ Ryan Deem _____ __01/26/25__

Team Member Date

_____ Zeynep Deniz Birgili _____ __01/26/25__

Team Member Date

_____ Austin Cook _____ __01/26/25__

Team Member Date